# MIN-JONG BONG

Ph.D. Candidate in Inorganic Chemistry · Korea University · Advisor: Prof. Ho-Jin Son

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## RESEARCH INTEREST

Ph.D. candidate in Inorganic Chemistry at Korea University (expected February 2027), specializing in the design of single-molecule Mn(I) catalysts for selective $CO_2$ reduction. Research centers on secondary coordination sphere engineering of well-defined molecular complexes — inspired by formate dehydrogenase enzymes — integrating experimental (operando FTIR, CV, XRD) and computational (DFT; ORCA) approaches with Python-based automation of computational workflows. Key results include a TON of ~300 with >94% formate selectivity, published in Chemical Science (2026, first author).

**Keywords:** Organometallic Synthesis, $CO_2$ Reduction, Photocatalysis, Secondary Coordination Sphere, Operando FTIR, DFT

## EDUCATION

**Ph.D. candidate, Inorganic Chemistry** — Mar 2024 – Present
Korea University · Department of Advanced Materials Chemistry · Advisor: Prof. Ho-Jin Son

**M.S., Inorganic Chemistry** — Mar 2022 – Feb 2024
Korea University · Department of Advanced Materials Chemistry · Advisor: Prof. Ho-Jin Son

**B.S., Advanced Materials Chemistry** — Mar 2018 – Feb 2022
Korea University (Sejong)

## PUBLICATIONS

[1] **Secondary Sphere Impact on Organometallic Catalysts in Photochemical $CO_2$ Reduction: A Comparative Study of Tethering Ligand Lengths with Secondary Coordination**
**Min-Jong Bong**, Wonjung Lee, Junhyeok Seo*, Ho-Jin Son*
In preparation **first author**

[2] **Unveiling Key Factors Governing the Catalytic Performance of Half-Metallocene Complexes for Photochemical $CO_2$ Reduction and Hydrogen Production: Relativistic Effects and Charge Delocalization**
Daehan Lee†, **Min-Jong Bong**†, Seunghwan Cha, Sangheon Jeong, Jung-Min Ji, Won-Jo Jung, Nady A. Fathy, Ho-Jin Son*
*Inorganic Chemistry Frontiers* · 2026 · *Royal Society of Chemistry* · DOI: 10.1039/D6QI01142J **co-first author †**

[3] **A secondary-sphere proton channel accelerating metal–hydride formation in Mn(I) catalysts for selective $CO_2$-to-formate conversion**
**Min-Jong Bong**†, Wonjung Lee†, Daehan Lee, Hyunuk Kim, Junhyeok Seo*, Ho-Jin Son*
*Chemical Science*, 17 (19), 9535–9551 · 2026 · *Royal Society of Chemistry* · DOI: 10.1039/D5SC09412G **first author**

[4] **Efficient and Durable Photochemical $CO_2$ Reduction by $TiO_2$-Immobilized Metal Porphyrin Catalysts**
Hyeongu Kang, Daehan Lee, Sangheon Jeong, Seunghwan Cha, **Min-Jong Bong**, Myung Jae Lee, Won-Jo Jung, Ho-Jin Son*
*ChemSusChem*, 19, e202501889 · 2026 · *Wiley-VCH / Chemistry Europe* · DOI: 10.1002/cssc.202501889

[5] **Organic phosphorescence in Pt(II)-complexes linked to organic chromophores for blue-emitting organic light-emitting diodes**
Seong Woon Jeong†, Hyung Joo Lee†, Bum June Park, **Min-Jong Bong**, Daehan Lee, Taekyung Kim*, Chul Hoon Kim*, Ho-Jin Son*
*Communications Chemistry*, 8, 140 · 2025 · *Nature Portfolio* · DOI: 10.1038/s42004-025-01533-y

## RESEARCH PROJECTS

### Tether Length Modulation of Secondary Coordination Layer
2024 – Present

Systematic variation of –OH/–OMe pendant length at 6,6′-bipy positions of Mn(I) complexes. Best performer Mn-DiMeOMe: TON 368, formate selectivity 97.8%.
Photocatalysis · CV · Operando FTIR · DFT · XRD

### Proton Relay Engineering in Mn(I) Complexes
2022 – 2025

Ethylene-bridged pendants mimic formate dehydrogenase. Mn-bpydiOMe: TON ~300, formate selectivity >94%. Published in Chem. Sci. 2026.
$CO_2$ Reduction · Operando FTIR · DFT · Mechanistic Study

## SKILLS & METHODS

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|---|---|
| **Technical:** | Organic & organometallic synthesis, Glovebox & Schlenk techniques, Air-sensitive compound handling, Photocatalysis, Sublimation purification, 3D Graphic Design (Adobe Dimension) |
| **Instrumental:** | Operando FTIR, Cyclic Voltammetry, UV-Vis Spectroscopy, Quantum Yield Measurement, Single-crystal XRD, Structure refinement (SHELXTL/SHELXL), NMR, ESI-MS, GC-MS, GC-TCD, HPLC |
| **Computational:** | DFT (ORCA, Gaussian), CPCM solvation, Reaction pathway mapping, Python (computational data automation & visualization), AI-assisted computational workflow (Claude Code) |

## TEACHING & MENTORING

### General Chemistry Laboratory 1 — Lecturer
Spring 2026

Serving as course instructor for general chemistry laboratory.

### Inorganic Chemistry Laboratory — Lecturer
Fall 2026

Serving as course instructor for inorganic chemistry laboratory.

### CURT Undergraduate Research — Mentoring
Feb – Jul 2024

Mentored undergraduates in research design and execution.

### General Chemistry Laboratory 1 — TA
Spring 2022, 2023, 2024, 2025

Supervised lab sessions for general chemistry.

### General Chemistry Laboratory 2 — TA
Fall 2022, 2024

Supervised lab sessions for general chemistry.

### Inorganic Chemistry Laboratory — TA
Fall 2023, 2024, 2025

Instructed inorganic synthesis experiments.

## HONORS & AWARDS

**Impressive Poster Award** — 2026
14th Asian and Oceanian Photochemistry Conference (APC 2026)
*A Secondary-Sphere Proton Channel Accelerating Metal–Hydride Formation in Mn(I) Catalysts for Selective $CO_2$-to-Formate Conversion*

**Excellence in Poster Presentation Award** — 2025
Inorganic Chemistry Division, Korean Chemical Society
*Second Coordination Sphere Engineering for Efficient and Selective $CO_2$-to-Formate Conversion of Mn(I) Bipyridyl Complexes: Steric and Brønsted Acidity/Basicity Influence of Alcohol and Alkyloxy Pendants and Its Mechanistic Investigation*

**Excellent Research Award** — 2021
CURT Research Program, Korea University
*A Hybrid Ru/$TiO_2$ Catalyst for Photocatalytic $CO_2$ to CO/Formate Conversion*